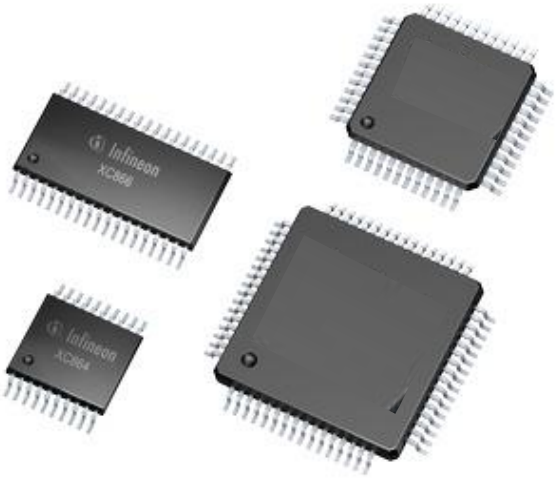
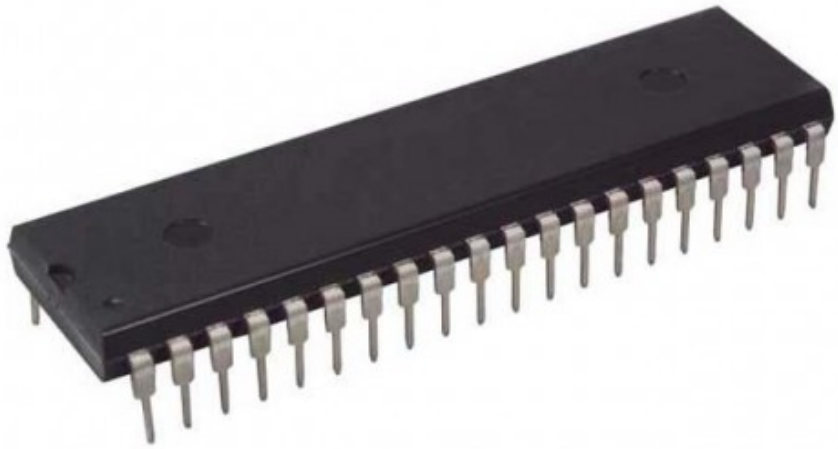
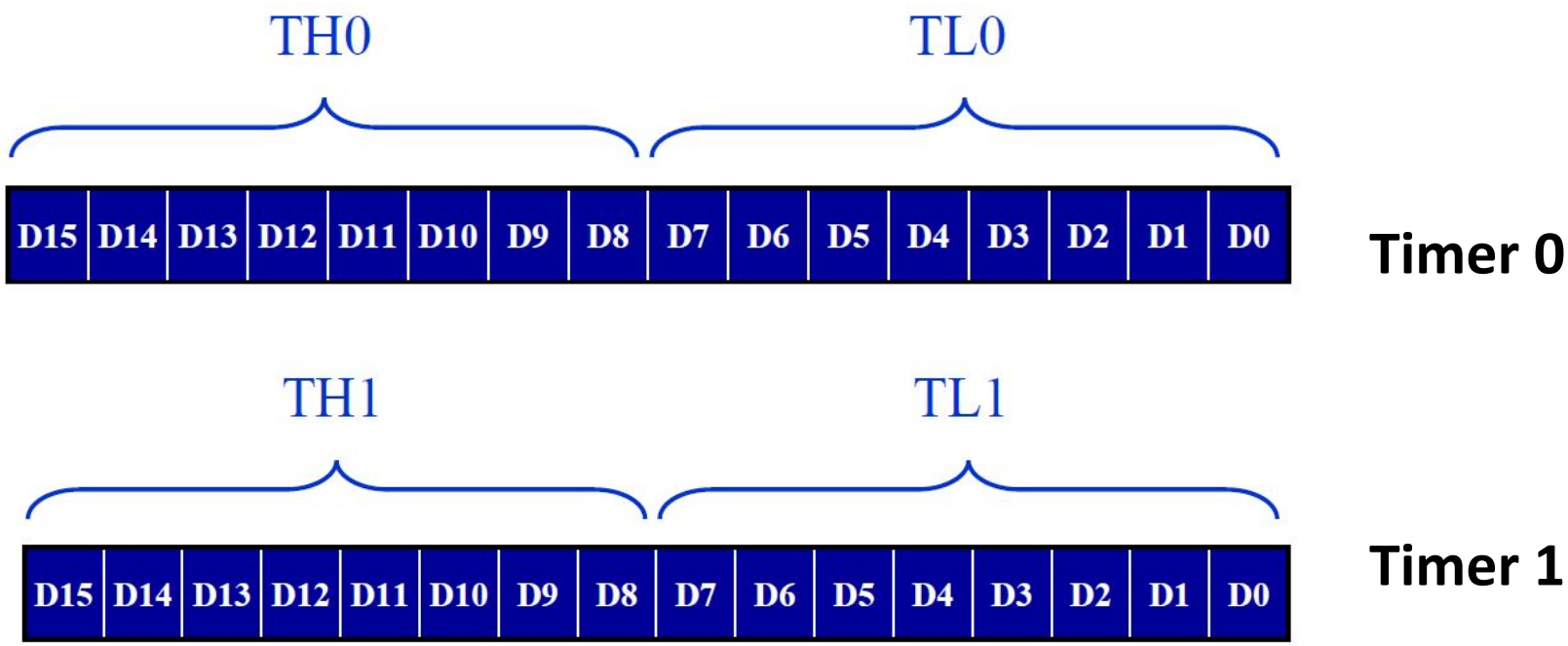
8051 Microcontroller



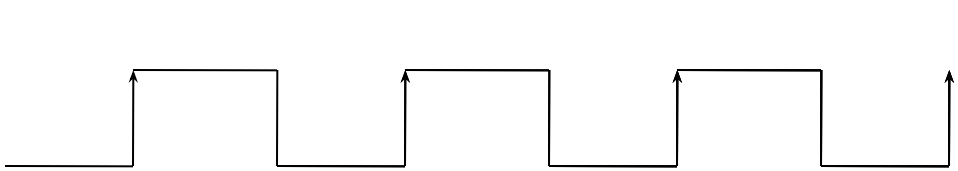


Timer in 8051

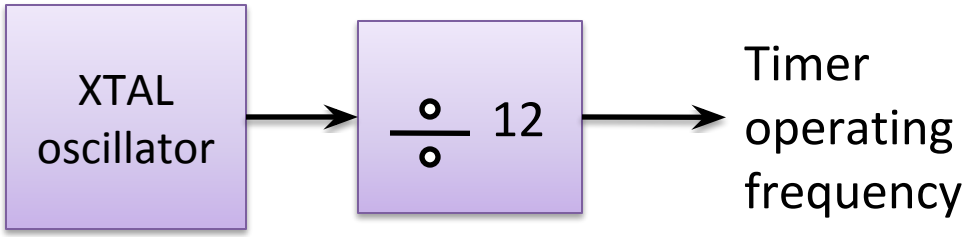
- The 8051 has two timers/counters
- **Timer-** to generate a time delay
- **Counter-**to count events happening outside the microcontroller
- Timer 0 and timer 1 are 16 bit



Every timer needs a clock pulse to tick



TH0	TL0
00H	03H



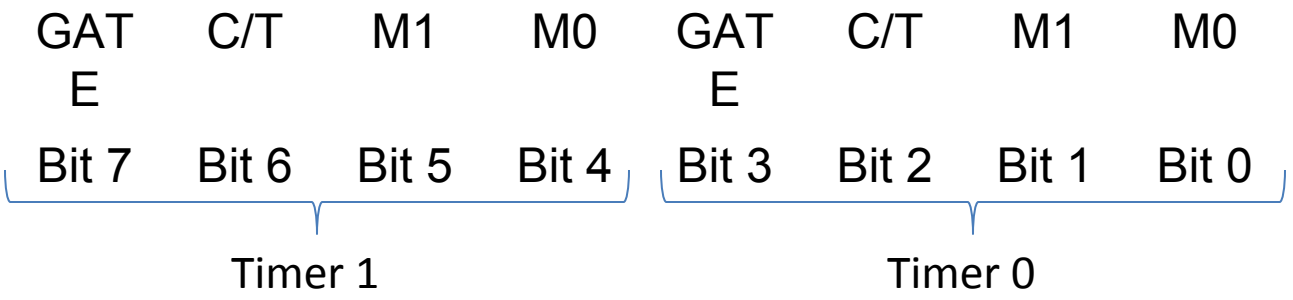
Timer frequency= Crystal frequency / 12

Crystal frequency= 11.0592 MHz

Then timer frequency= 1.085 us

TMOD: timer/Counter mode control register

- TMOD is a 8-bit register
- Use to set the various timer operation modes

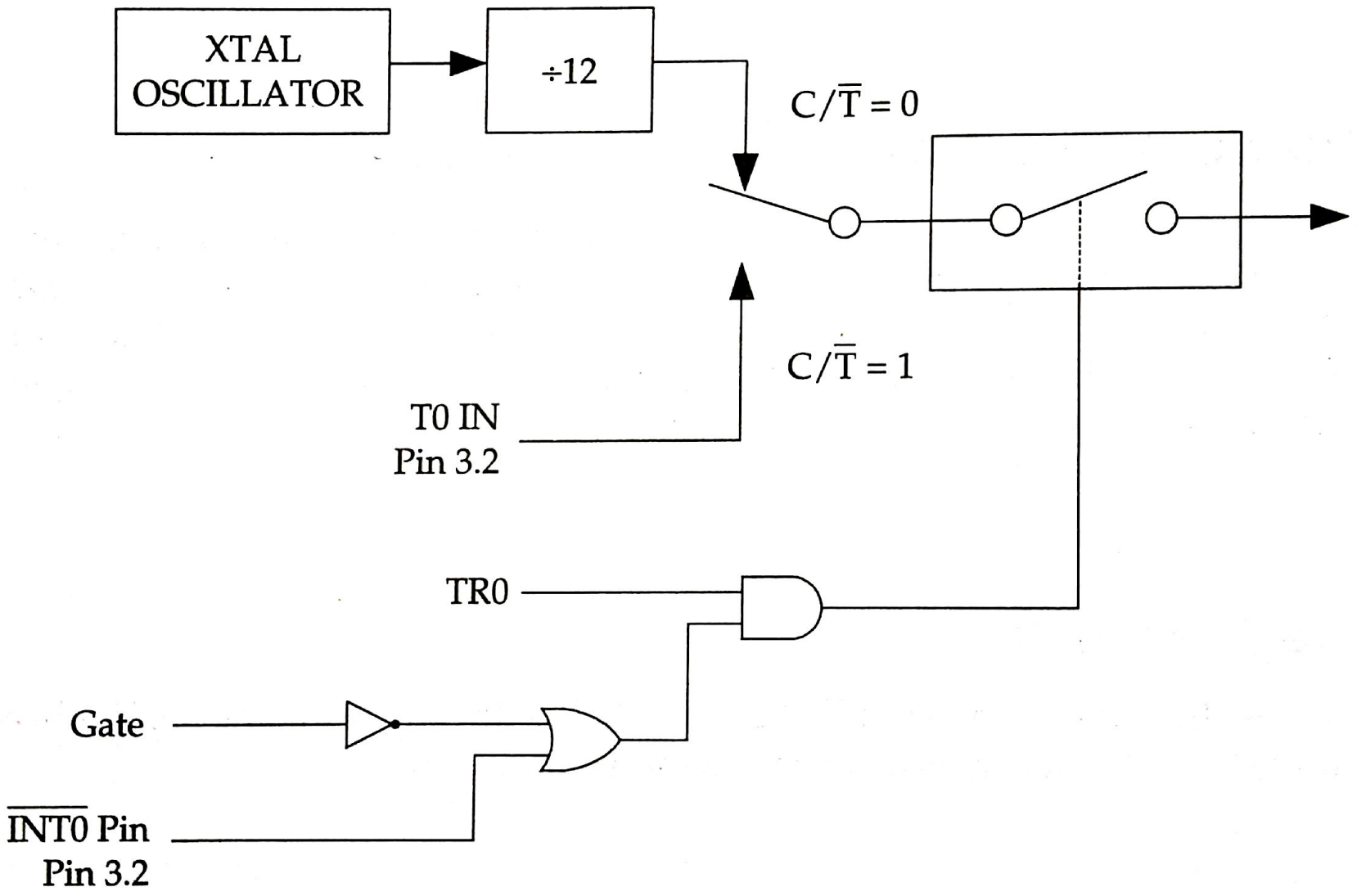


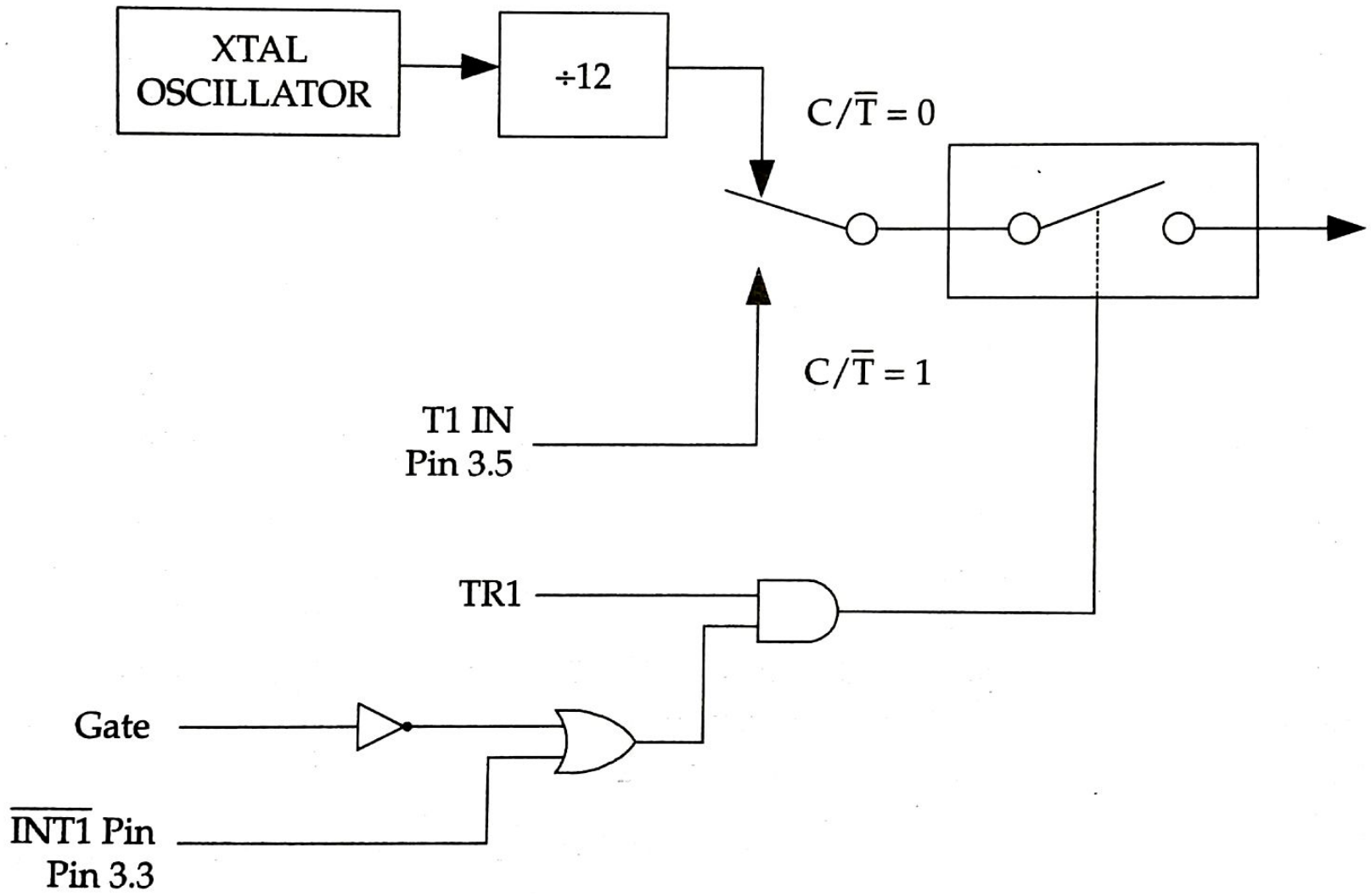
M1	M0	Mode	Operating mode
0	0	0	13-bit timer mode 8-bit timer/counter THx with TLx as 5-bit prescaler
0	1	1	16-bit timer mode 16-bit timer/counter THx and TLx are cascaded; there is no prescaler
1	0	2	8-bit auto reload 8-bit auto reload timer/counter; THx holds a value which is to be reloaded TLx each time it overfolws
1	1	3	Split timer mode

TMOD: timer/Counter mode control register

GATE	C/T	M1	M0	GATE	C/T	M1	M0
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0

Bit	Operating mode
C/T	C/T = 1, counter C/T =0, timer
GATE	GATE=0, TRX=1, timer on GATE=1, INTX=1, TRX=1, timer on


XTAL
OSCILLATOR
÷12
C/T = 0
C/T = 1
T0 IN
Pin 3.2
TR0
Gate
INT0 Pin
Pin 3.2


XTAL
OSCILLATOR
÷12
C/T = 0
C/T = 1
T1 IN
Pin 3.5
TR1
Gate
INT1 Pin
Pin 3.3

TCON: Timer/ Counter control register

TF1	TR1	TF0	TR0	IE1	IT1	IE0	IT0
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0

TF1	Timer 1 overflow flag
TR1	Timer 1 run control bit
TF0	Timer 0 overflow flag
TR0	Timer 0 run control bit
IE1	interrupt
IT1	interrupt
IE0	interrupt
IT0	interrupt

TMOD	GATE	C/T	M1	M0	GATE	C/T	M1	M0
	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
TCON	TF1	TR1	TF0	TR0	IE1	IT1	IE0	IT0
	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0

TMOD reg. is not bit addressable

Ex. MOV TMOD, #02H

TCON reg. is bit addressable

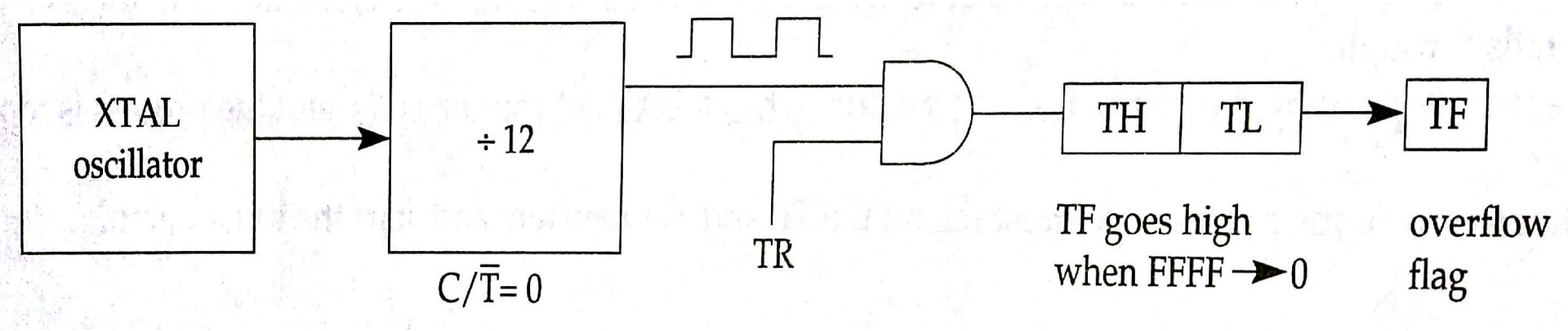
Ex. SETB TR1

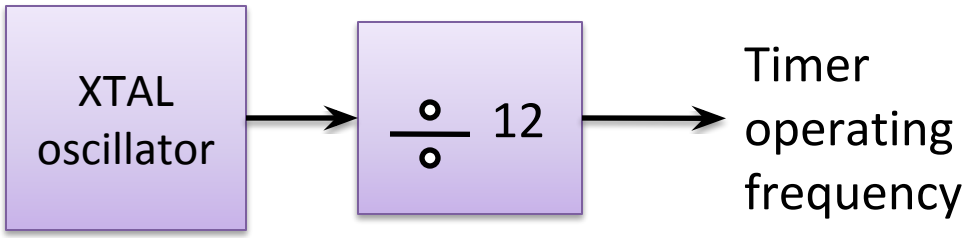
THx and TLx can be use like general purpose register

Ex. MOV TH0, #05H

Mode 1

- 16 bit timer
- 0000h to FFFFh
- After the timer is started, it start to count up. It count up until it reaches its limit of FFFFH.
- When timer rolls over from FFFFH to 0000H , it sets high a flag bit called TF(timer flag)





Timer frequency= Crystal frequency / 12
Crystal frequency= 11.0592 MHz
Then timer frequency= 921.6 KHz
i.e each timer clock has period of (1/921.6 KHz)=1.085us



TH0	TL0
00H	03H

1.085 us

1.085 us x 2

1.085 us x 3

1 count = 1.085 us

No. of Count $\propto$ Delay

Calculate a values of THx and TLx for generation of 5 ms delay ?

Timer generated Delay = (65536 - NNNN) x 1.085 us

NNNN = 65536 - (Timer generated Delay / 1.085 us)

XXXX= (NNNN)H

THx	TLx	F F	FF
THx	**TLx**	**E** **E**	**00**

If timer is 16 bit, what will be the maximum delay the timer can generate ?

TH0 TL0 F FF
F

TH0 TL0 0 00
0

The number of counts for rollover is FFFF-0000= FFFF(65535 decimal)
One extra clock needed when timer roll over from FFFF to 0 and rises TF flag.

Maximum delay= (65535+1) x 1.085 us =71.1 ms

Timer generated Delay = (FFFF - xxxx + 1) x 1.085 us

LED blinking

TMOD	GATE	C/T	M1	M0	GATE	C/T	M1	M0
	0	0	0	0	0	0	0	1

	MOV TMOD, #01H	Timer 0, mode 1, 16 bit mode
Main code	**Again1:**MOV P1, #0FFH	Turn on Led
	ACALL **DELAY**	Call delay
	MOV P1, #00H	Turn off Led
	ACALL **DELAY**	Call delay
	SJMP **Again1**	Short jump to L1
	DELAY: MOV TL0, #00H	TL0=00h
	MOV TH0, #0EEH	TH0=EEh
Sub-routine	SETB TR0	Turn on timer 0
	again: JNB TF0 again	Monitor TF0 flag until it rolls over
	CLR TR0	Stop timer
	CLR TF0	Clear overflow flag
	RET	Return from subroutine

```
ORG 00H

MOV  TMOD, #01H
again1:
MOV P1, #0FFH
ACALL DELAY
MOV P1, #00H
ACALL DELAY
SJMP again1

DELAY:    MOV TL0, #00H
MOV TH0, #0EEH
SETB TR0
again: JNB TF0, again
CLR TR0
CLR TF0
RET


END
```

Calculate a values of THx and TLx for generation of 1 sec delay ?

NNNN = 65536 - (Timer generated Delay / 1.085 us)

1 sec= 200 x 5 ms

NNNN= 65536 - (5ms/ 1.085 us) = 60928 = (EE00)h



THx TLx FF FF

X 200

THx TLx EE 00



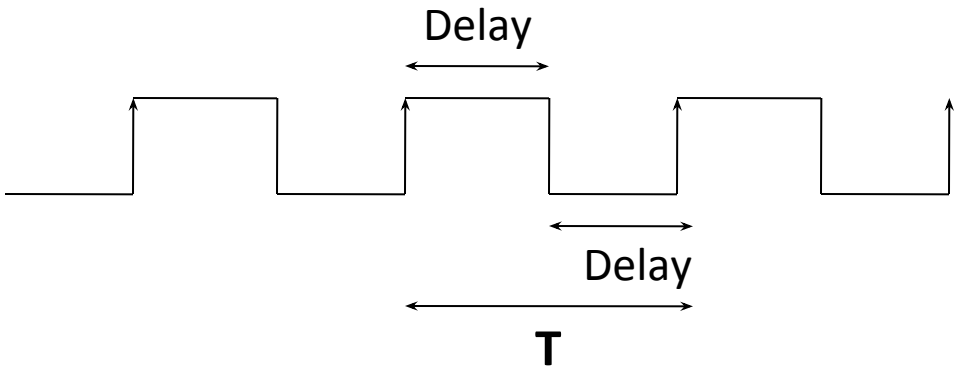
LED blinking

Main code

Sub-routine

MOV TMOD, #01H	Timer 0, mode 1, 16 bit mode
again1: MOV P1, #0FFH	Turn on Led
ACALL **DELAY**	Call delay
MOV P1, #00H	Turn off Led
ACALL **DELAY**	Call delay
SJMP **again1**	Short jump to L1
DELAY: MOV R0,#20	Set count value 20
Back: MOV TL0, #00H	TL0=00h
MOV TH0, #0EEH	TH0=EEh
SETB TR0	Turn on timer 0
again: JNB TF0 again	Monitor TF0 flag until it rolls over
CLR TR0	Stop timer
CLR TF0	Clear overflow flag
DJNZ R0, **Back**	
RET	Return from subroutine

Generation of square wave ?



F = 1 / T

Find the value of THx and TLx for generation a frequency 100 KHz ?

Square wave Frequency = 100 KHz

Square wave Time period = 10 us

Delay = (10 us / 2) = 5 us

NNNN = 65536 - (Timer generated Delay / 1.085 us)

XXXX= (NNNN)H

= 65536-(5 us / 1.085 us)= 65531=(FFFB)h

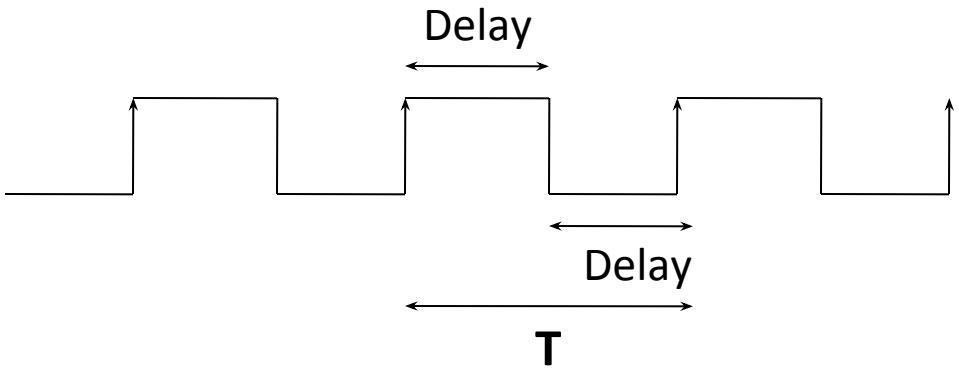
TH0	TL0	F	FF
		F	

TH0	**TL0**	**F**	**F**
		F	**B**

- In order to repeat the process the register THx and TLx must be reloaded with orignal values,

- TFx must be reset to 0

Minimum frequency that 8051 can generate ?



$$F = 1 / T$$

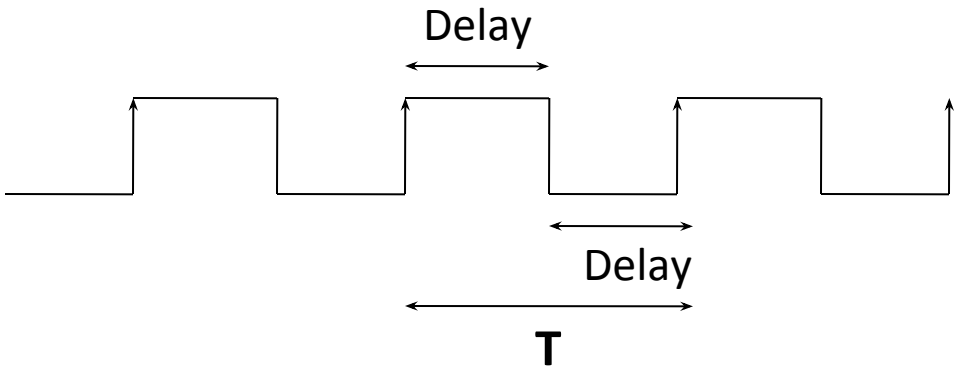
Minimum frequency means maximum time period

Maximum time period = 71 ms

Therefore minimum frequency = 1 / (2 x 71 ms) = 7 Hz

TH0	TL0	F	FF
		F	
TH0	**TL0**	**00**	**00**

Maximum frequency that 8051 can generate ?



F = 1 / T

Minimum frequency means maximum time period

minimum time period = 1.085 us

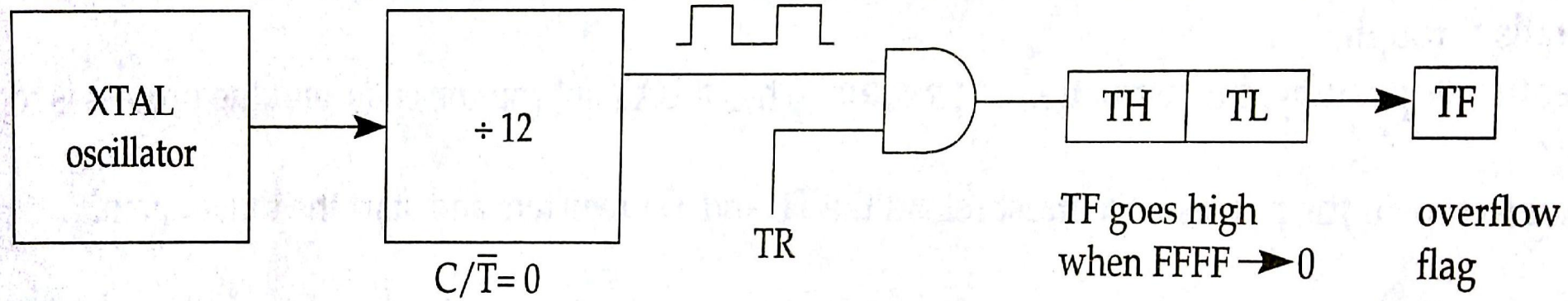
Therefore maximum frequency = 1/(2 x 1.085 us) = 461 KHz

TH0	TL0	F	FF
		F	

TH0	**TL0**	**F**	**F**
		F	**B**

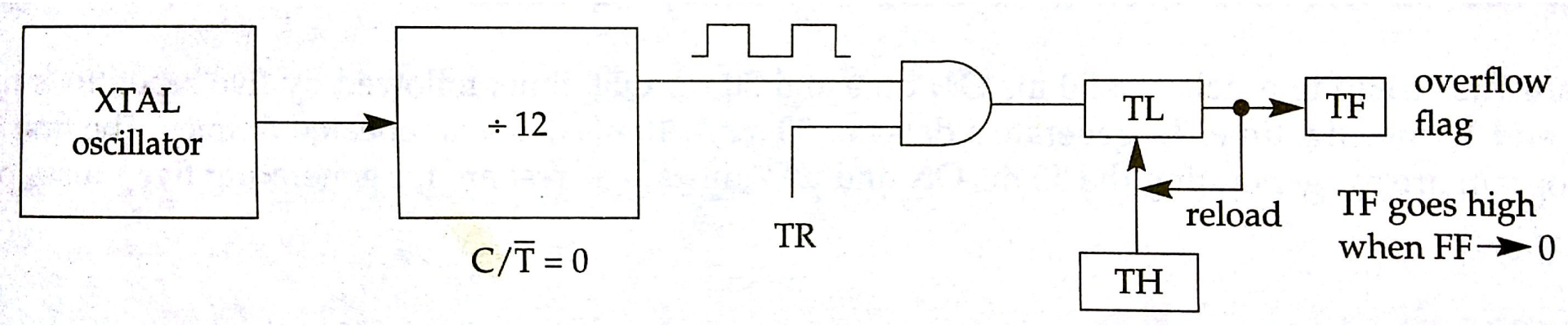
Mode 0

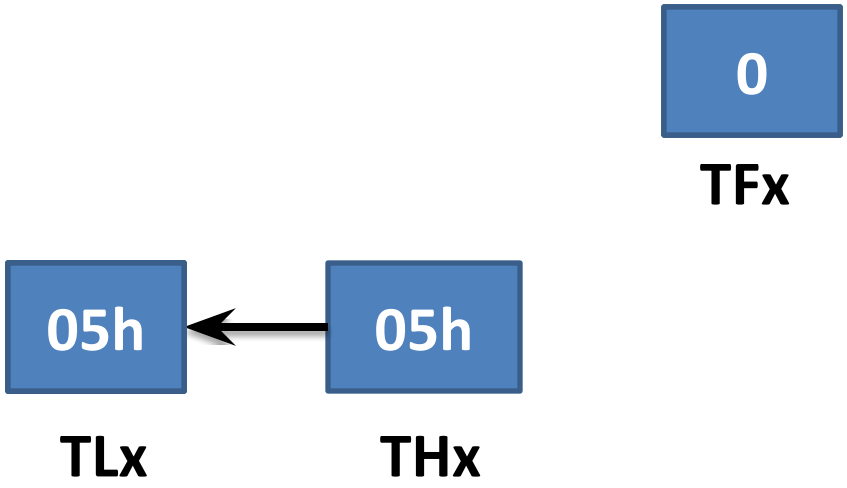
- 13 bit timer
- 0000h to 1FFFh
- After the timer is started, it start to count up. It count up until it reaches its limit of 1FFFH.
- When timer rolls over from 1FFFH to 0000H , it sets high a flag bit called TF(timer flag)
- Mode 0 is exactly like mode1 except that it is a 13 bit timer instead of 16 bit.



Mode 2

- 8 bit timer, auto reload mode
- 00h to FFh
- After the timer is started, it start to count up. It count up until it reaches its limit of FFH.
- When timer rolls over from FFH to 00H , it sets high a flag bit called TF(timer flag)
- When TLx reg. roll from FFh to 00h and TFx set to 1, TL reloaded automatically by original value kept by THx register.




0
TFx
05h
05h
TLx
THx

Calculate a maximum delay that can generate in mode 2 ?

Timer generated Delay = (256 - NN) x 1.085 us

NN = 256 - (Timer generated Delay / 1.085 us)
XX= (NN)H

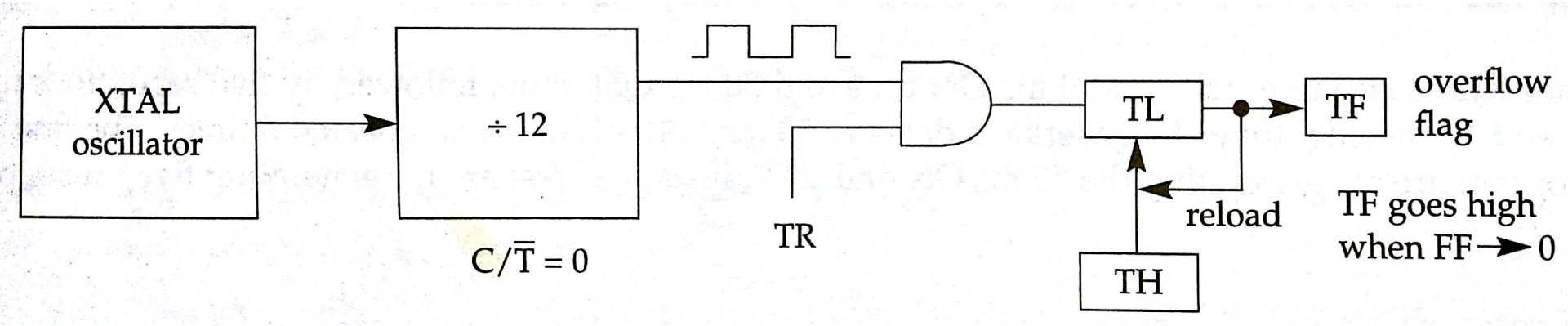
Max delay= 256 x 1.085 us= 0.277 ms

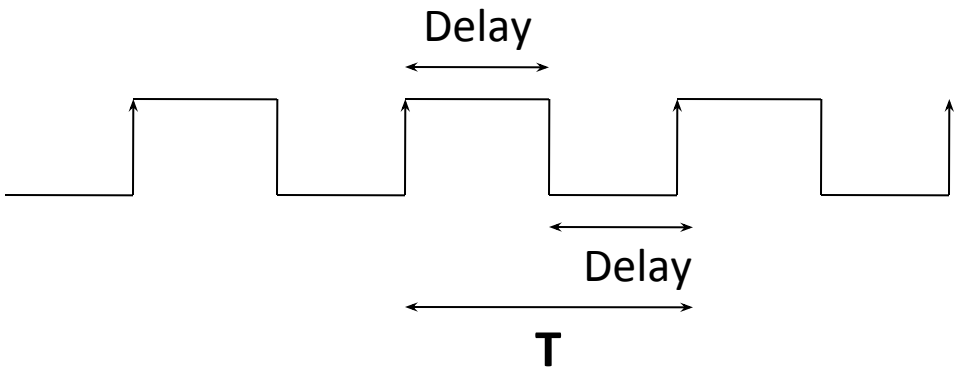
Minimum freq= 1/ (2 x 256 x 1.085 us) = 1.8 KHz

THx FF

THx 00

Generate square wave of 1 KHz on pin P1.2 using XTAL=22MHz





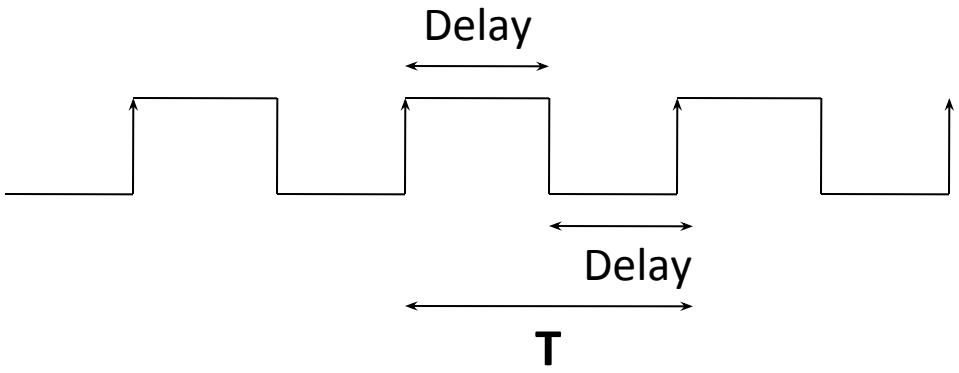
F = 1 / T

Square wave Frequency = 1 KHz

Square wave Time period = 1 ms

Delay = (10ms / 2) = 0.5 ms

Generation of square wave ?



F = 1 / T

1 count = 1/ timer frequency

= 1/(XTAL/12)

= 1/(22Mhz/12)

= 0.546 us

TH0	TL0	F	FF
		F	

TH0	**TL0**	**F**	**F**
		F	**B**

1 count= 0.546 us
X count = 0.5 ms
X= 916

Note: X > 256 (max count)

1 count = 1/ timer frequency

= 1/(XTAL/12)

= 1/(22Mhz/12)

= 0.546 us

1 count= 0.546 us
X count = 0.5 ms
X= 916

Note: X > 256 (max count)

1 count= 0.546 us
X count = 100 us x 5

1 count= 0.546 us
X count = 100 us

X= 183

NN = 256 - (Timer generated Delay / 1.085 us)
XX= (NN)H

NN = 256 - (Timer generated Delay / 0.546 us)
= 256-183
=73
=49H

X 5

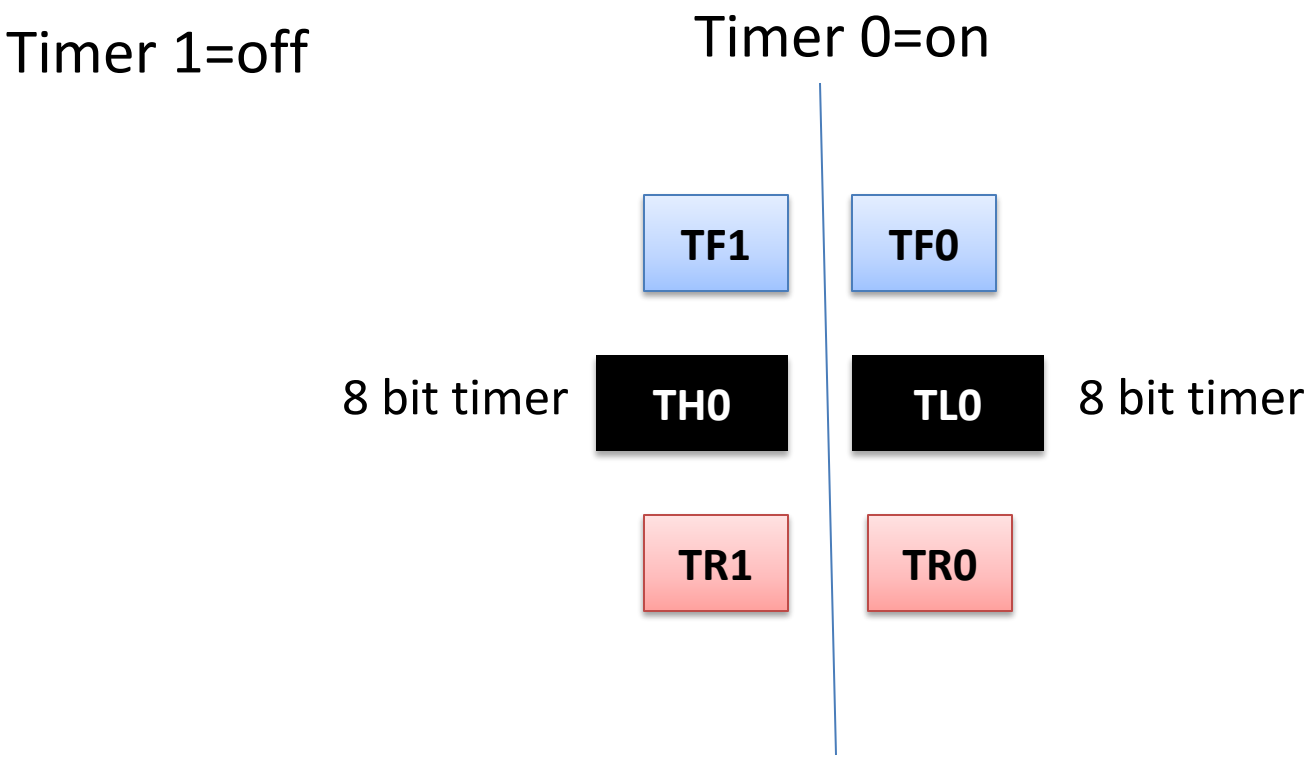
LED blinking

TMOD	GATE	C/T	M1	M0	GATE	C/T	M1	M0
	0	0	1	0	0	0	0	0

	MOV TMOD, #20H	Timer 1, mode 2, 8 bit mode
Main code	**Again1:**MOV P1, #0FFH	Turn on Led
	ACALL **DELAY**	Call delay
	MOV P1, #00H	Turn off Led
	ACALL **DELAY**	Call delay
	SJMP **Again1**	Short jump to L1
Sub-routine	**DELAY:** MOV R0, #05	
	HERE: MOV TH1, #49h	TH0=EEh
	SETB TR1	Turn on timer 0
	again: JNB TF1 again	Monitor TF0 flag until it rolls over
	CLR TR1	Stop timer
	CLR TF1	Clear overflow flag
	DJNZ R0,HERE	counter
	RET	Return from subroutine

Mode 3

- Split timer mode
- Timer 0 in Mode 3 establishes TL0 and TH0 as two separate 8-bit counters
  (Timer 0) TL0 is an 8-bit Timer/Counter controlled by the standard Timer 0 control bits.
  TH0 is an 8-bit timer only controlled by Timer 1 control bits.
- (Timer 1) Timer/Counter 1 stopped.



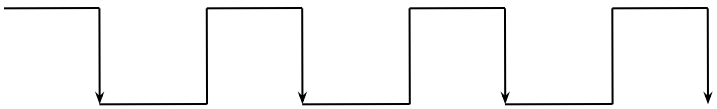
Counter

- Every thing is same as that of timer, the only difference is clock source.

TMOD	GATE	C/T	M1	M0	GATE	C/T	M1	M0
		1				1		

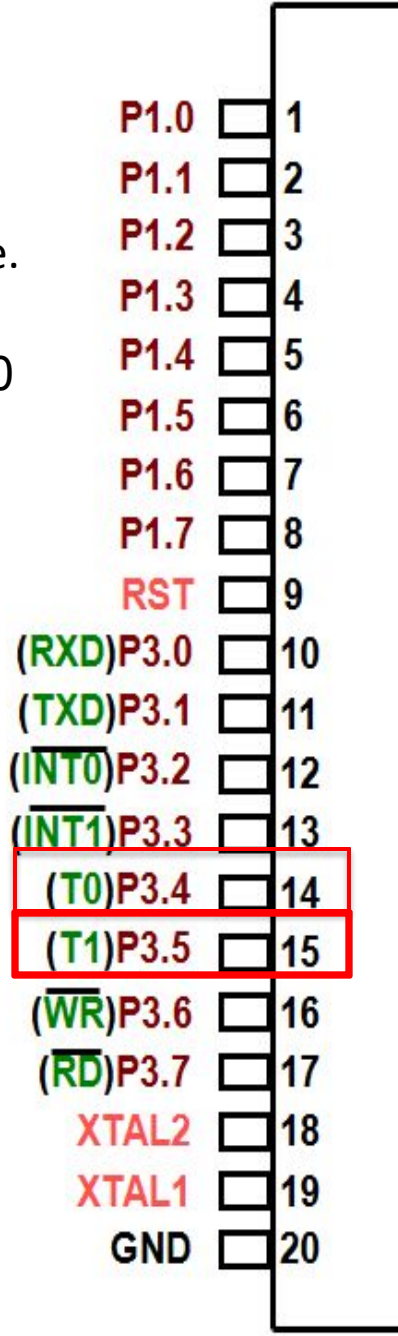
In the 'Counter' function, the register is incremented in response to a **1-to-0 transition** at its corresponding external input pin, T0 or T1. In this function, the external input is sampled once every machine cycle.

External clock pulse

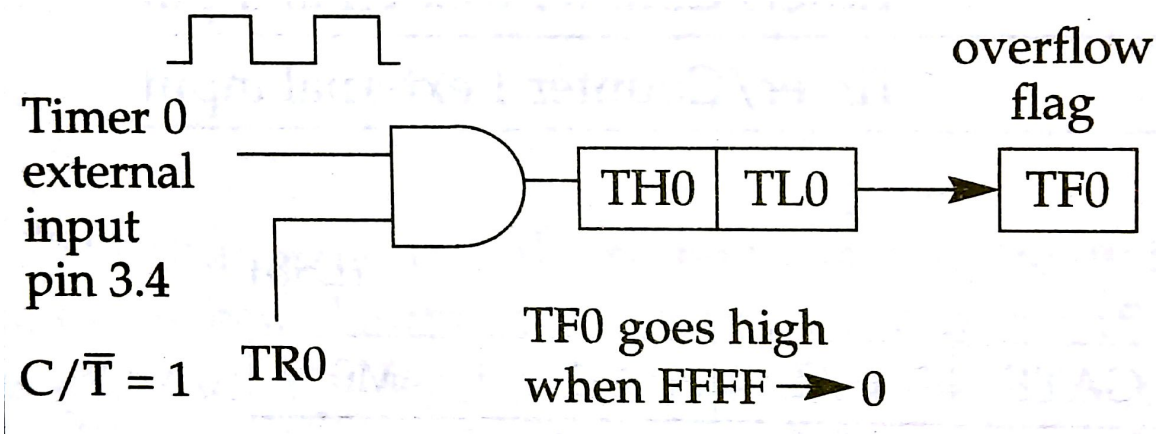


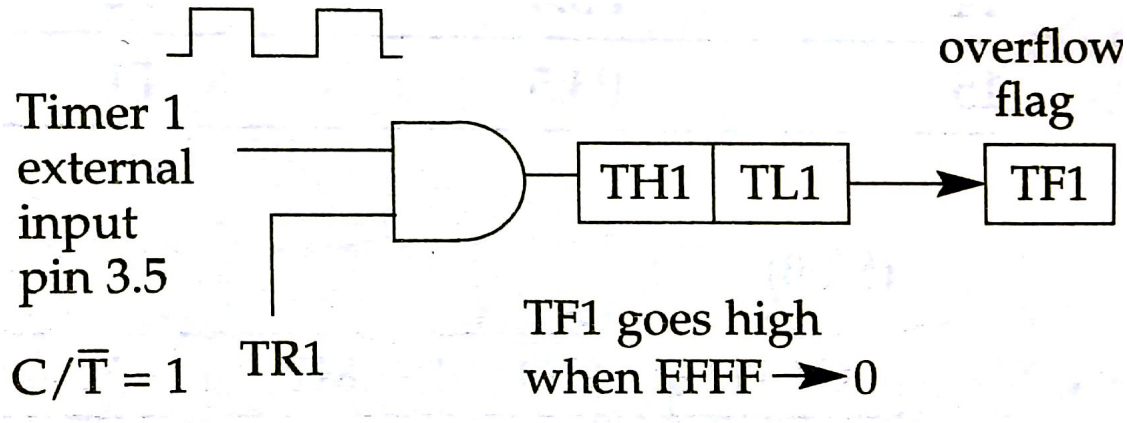
Counter 0

Counter 1

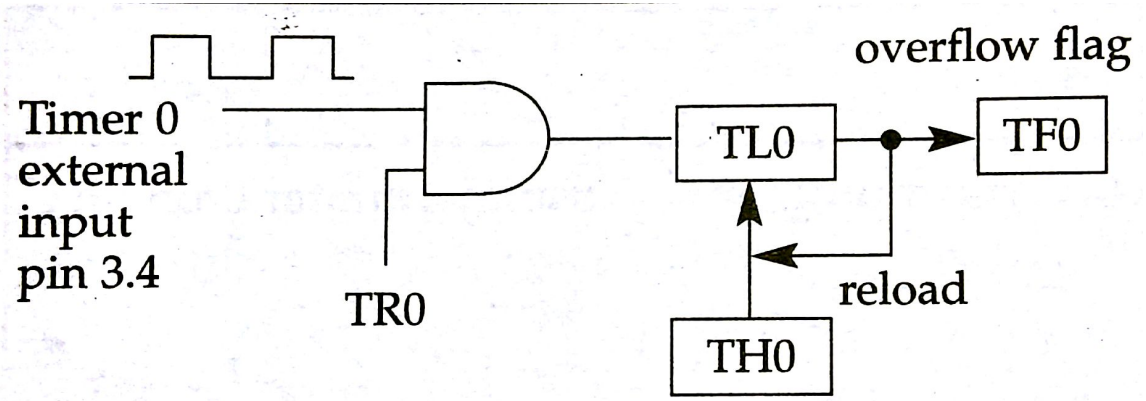


Mode 1


Timer 0
external
input
pin 3.4
C/T = 1
TR0
TH0
TL0
TF0
overflow
flag
TF0 goes high
when FFFF → 0

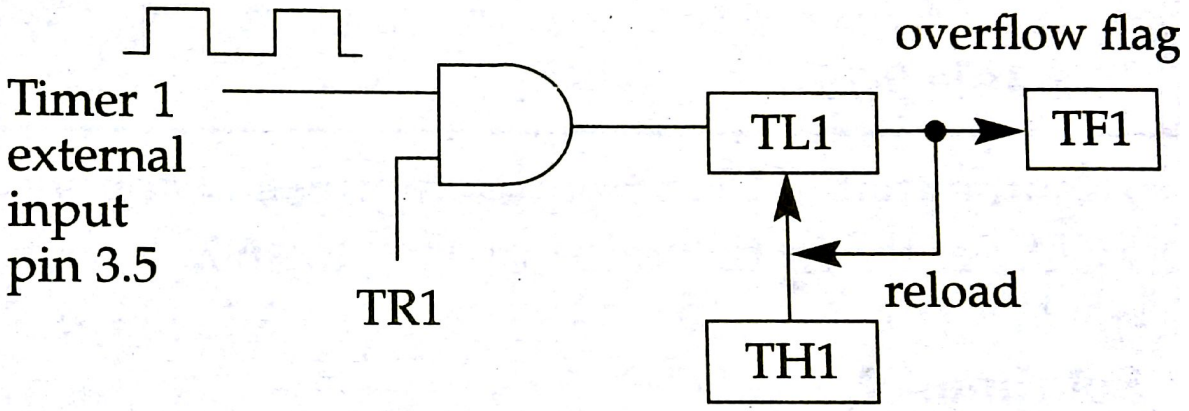

Timer 1
external
input
pin 3.5
C/T = 1
TR1
TH1
TL1
TF1
overflow
flag
TF1 goes high
when FFFF → 0

Mode 2



$C/\overline{T} = 1$

TF0 goes high
when FF → 0



$C/\overline{T} = 1$

TF1 goes high
when FF → 0

TMOD	GATE	C/T	M1	M0	GATE	C/T	M1	M0
	0	0	0	0	0	0	0	1

Toggle led after 5 time switch press

```
ORG 00H

MOV  TMOD, #50H              ; COUNTER1 MODE 1
HERE:

MOV TH1,#0FFH                ; TO COUNT 5
MOV TL1,#251                 ; TO COUNT 5
SETB TR1                     ; TIMER 1 ON
again: JNB TF1, again        ; WAIT TILL TF1=1
CLR TR1                      ; STOP TIMER1
CLR TF1                      ; CLEAR TF FLAG
CPL P1.2                     ; COMPLEMENT P1.2

SJMP HERE

END
```